

PROPOSAL DOCUMENTATION

Technical Requirements Specification

Sample specification for the proposed application portal based on the current demo, stakeholder workflow requirements, and the expected production operating model.

Project	Application Portal
Document Type	Technical Requirements
Revision	Draft v1.0
Date	29 April 2026

This specification is intended for stakeholder review and planning. Final production details should be confirmed during discovery.

REQUIREMENTS

1. Purpose and Scope

The system will support an annual application programme that includes applicant registration, eligibility screening, document submission, multi-stage review, comparative scoring, interview scoring, final decisions, and applicant communications.

2. In Scope

- Applicant-facing portal with required forms and document upload
- Multi-role staff dashboard with workflow-gated access
- Essay, academic, document, shortlisting, interview, and final decision stages
- Master scorecard with ranking and export capability
- Audit logging, encryption, data export, and deletion workflows

3. Out of Scope for Initial Demo

- Production authentication integration
- Live payment processing
- External student information system integration
- Bulk data migration from legacy platforms

REQUIREMENTS

4. Functional Requirements

4.1 Applicant Portal

ID	Requirement
FR-01	The system shall present three initial eligibility questions before the main application form.
FR-02	If any eligibility answer is negative, the system shall stop the application flow and display a clear ineligibility message.
FR-03	The system shall reserve an application reference number even for ineligible submissions for reporting purposes.
FR-04	The system shall capture explicit consent before personal information is submitted.
FR-05	The system shall require upload of mandatory supporting documents before final submission.
FR-06	The system shall prevent duplicate active submissions for the same applicant within the same cycle.

4.2 Review Workflow

ID	Requirement
FR-07	Essay reviewers shall score essays without visibility into academic results or peer reviewer scores.
FR-08	Academic reviewers shall capture subject marks and the system shall calculate the average automatically.
FR-09	Document checkers shall record required document completeness and issue notes.
FR-10	Screeners shall access aggregated score and checklist views only after prior stages are complete.

FR-11	Interview admins shall capture manual interview scores and notes for shortlisted applicants.
FR-12	Final decision makers shall assign Approved, Rejected, or Standby outcomes.

REQUIREMENTS

5. Workflow and Data Rules

5.1 Workflow Enforcement

- The system shall enforce ordered stage progression.
- The system shall prevent shortlisting until essay review, academic scoring, and document verification are complete.
- The system shall prevent interview scoring unless the applicant has been shortlisted.
- The system shall prevent final decision completion until interview scoring is complete where required.

5.2 Scorecard Rules

- Teaching experience scoring shall be rule-driven, for example 3+ years = 5 points, 1-2 years = 3 points, under 1 year = 1 point.
- TEFL and degree presence shall contribute binary points as configured.
- Essay and interview scores shall be pulled from role-owned scoring records.
- The system shall calculate total score automatically and rank applicants by score.
- The system shall support filtering and spreadsheet export of scorecard data.

6. Role and Permission Requirements

ID	Requirement
PR-01	Every user shall be assigned one or more roles and all access shall be policy-controlled.
PR-02	The API shall enforce role permissions independently of the user interface.
PR-03	Users shall view only the applicant fields required for their role.
PR-04	Reviewer scores from other users shall remain hidden until authorized senior stages.
PR-05	Administrative actions affecting roles or permissions shall be fully audited.

REQUIREMENTS

7. Security, POPIA, and Operations

7.1 POPIA and Security Requirements

ID	Requirement
SR-01	All personal data shall be encrypted in transit and at rest.
SR-02	The system shall maintain an immutable audit trail of access and changes.
SR-03	The system shall support applicant data export requests.
SR-04	The system shall support deletion or anonymization workflows after the application cycle, subject to retention policy.
SR-05	The system shall support breach response and notification procedures.
SR-06	Passwords shall be stored using secure hashing such as Argon2id or bcrypt.
SR-07	Sessions shall expire after inactivity and shall be rotated on sensitive events.
SR-08	The application shall be delivered over HTTPS only.

7.2 Audit Logging

Audit records shall capture, at minimum:

- actor user ID and active role
- target entity or application ID
- action performed
- timestamp in UTC
- before and after change summary where applicable
- IP address, user agent, and correlation ID

8. Non-Functional Requirements

Category	Requirement
Availability	The platform should be available throughout the application cycle with planned maintenance windows communicated in advance.
Performance	Normal page loads should remain responsive under expected seasonal usage and staff review operations.
Scalability	The solution should scale to support increased applicant volume without redesign of the workflow model.
Maintainability	The codebase shall use a layered architecture and version-controlled migrations.
Observability	The system shall emit structured logs, alerts, and job status information for support and incident response.

REQUIREMENTS

9. Suggested Technical Stack

Application Stack

- Frontend: Next.js or a .NET-aligned web front end
- Backend: ASP.NET Core Web API on .NET 8
- Database: PostgreSQL
- Background processing: hosted services or job runner

AWS Services

- CloudFront and WAF
- App Runner or ECS Fargate
- RDS PostgreSQL
- S3, KMS, SES, CloudWatch

10. Deliverables

- source code repositories
- database schema and migrations
- deployment instructions
- administrator and user manuals
- security and architecture documentation

This document is intended as a formal sample requirements specification aligned with the current proposal stage.